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Assessment and process optimization of high throughput biofabrication of immunocompetent breast cancer model for drug screening applications

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Abstract

Recent advancements in 3D cancer modeling have significantly enhanced our ability to delve into the intricacies of carcinogenesis. Despite the pharmaceutical industry's substantial investment of both capital and time in the drug screening and development pipeline, a concerning trend persists: drug candidates screened on conventional cancer models exhibit a dismal success rate in clinical trials. One pivotal factor contributing to this discrepancy is the absence of drug testing on pathophysiologically biomimetic 3D cancer models during pre-clinical stages. Unfortunately, current manual methods of 3D cancer modeling, such as spheroids and organoids, suffer from limitations in reproducibility and scalability. In our study, we have meticulously developed 3D bioprinted breast cancer model utilizing decellularized adipose tissue-based hydrogel obtained via a detergent-free decellularization method. Our innovative printing techniques allows for rapid, high-throughput fabrication of 3D cancer models in a 96-well plate format, demonstrating unmatched scalability and reproducibility. Moreover, we have conducted extensive validation, showcasing the efficacy of our platform through drug screening assays involving two potent anti-cancer drugs, 5-Fluorouracil and PRIMA-1^{Met}. Notably, our platform facilitates effortless imaging and gene expression analysis, streamlining the evaluation process. In a bid to enhance the relevance of our cancer model, we have introduced a heterogeneous cell population into the DAT-based bioink. Through meticulous optimization and characterization, we have successfully developed a biomimetic immunocompetent breast cancer model, complete with microenvironmental cues and diverse cell populations. This breakthrough paves the way for rapid multiplex drug screening and the development of personalized cancer models, marking a paradigm shift in cancer research and pharmaceutical development.

1. Introduction

GLOBOCAN 2020 WHO reports indicate the incidence rates for breast cancer are at an all-time high and will continue to increase from 2.26 million new cases in 2020 to 3.19 million new cases in 2040, while the mortality rates will increase from nearly 685 thousand deaths in 2020 to approximately 1.04 million by the year 2024. Since cancer is an ever-evolving disease with cancer cells showing chemoresistance and residual cancer stem cells, the chemotherapeutic regimens constantly need to be updated and evaluated at the behest of clinicians and doctors handling patients. This calls attention to the FDA drug approval process for anti-cancer drugs, which is a substantial monetary investment. While millions of drug candidates are tested *in vitro*, most of these fail during the pre-clinical phase or clinical trials. The drug discovery and development pipeline has a timeframe of 10–15 years till a drug candidate hits the market, costing billions of dollars. It has been known to have a low success rate when it comes to anti-cancer drugs with a low rate of FDA approvals. A physiologically human-relevant drug screening platform that can bridge that gap between *in vitro* cancer models and animal models. They can help reduce the overall process's cost and remove the need for unnecessary animal sacrifice and time required for translational studies. Mutated p53 is seen in 50% of human cancers, emphasizing its role in tumorigenesis, and p53 mutations are reported in 60%–80% of cases of triple-negative breast cancer (TNBC) as one of the most common genetic alterations. Mutant p53 positivity is correlated with worse prognosis in TNBC patients [1–3]. Immunohistochemical WT p53 tumors were associated with better breast cancer-specific survival in initial N-stage patients [4]. Hence, developing potential anticancer therapies targeting mutant p53 with the least to no side effects is essential [5].

3D bioprinting has been extensively used to address research problems in tissue engineering and regenerative medicine and has been further adapted into 3D cancer modeling, allowing for the development of a physiologically relevant drug testing platform [6, 7]. 3D bioprinting can help recreate spatially complex heterogeneous tumors to create more relevant drug testing platforms. Modulating biomaterials and bioprinting modality can recreate different stages of carcinogenesis; complex microfluidics-based 3D cancer models can also be used to recreate dynamic tumor microenvironments. Biomaterial selection is essential for mimicking 3D tumor microenvironmental cues to develop a relevant model [8]. ECM-based hydrogel systems serve as an excellent source of natural biomaterials by providing native microarchitecture and biochemical cues [9]. Specifically, tissue decellularization can provide tissue-specific decellularized ECM hydrogel that can help recreate biomimetic tissue microenvironmental cues to study the pathophysiology of cancer [10], for example, to recapitulate glioblastoma microenvironment, decellularized brain ECM has been used [11], similarly decellularized liver has been used to study liver cancer [12–14]. Usually, dECM hydrogels are known to have a low viscosity and additive biomaterials have often been added to increase the viscosity or stiffness of the bioink and make it printable [15–18]. Another technique is hybrid 3D bioprinting, where the supporting framework was printed using PLA with dECM hydrogel stacked in between [19]. Novel crosslinking strategies have also been shown to produce a mechanically stable construct [21].

Adipose tissue forms an important structural part of mammary tissue [22, 23] and is known to help in mammary epithelialization [24, 25]. Previously, adipose tissue has been decellularized using physical, chemical, enzymatic methods, and a combination of all the methods [26]. Detergent and non-detergent-based decellularization methods of adipose tissue are also known [20, 27]. There is a range of adipose tissue decellularization techniques; however, they have multiple associated drawbacks like batch-to-batch variations, time-consuming protocols, loss of native ECM architecture due to harsh detergent washes and solvent treatments as well as leftover lipids in tissue that might hamper cytocompatibility at later stages [10, 26]. Decellularized adipose tissue (DAT) hydrogel-based 3D scaffolds have been previously used in the field of tissue engineering and regenerative medicine [28–40]; as well as for bioink formulation for extrusion-based 3D bioprinting [20, 41]. DAT hydrogel has been used as a 3D culture system to study carcinogenesis in three-dimension [42–47]. Similarly, DAT hydrogel-based cancer models have been fabricated to study breast cancer progression [48, 49]. However, the studies with breast cancer models involved bioink formulations with additive materials, making it hard to isolate the biological effect of dECM alone. Additionally, the monoculture printed scaffolds did not replicate the heterogeneous cell population within a tumor and did not allow for a high throughput testing system that could be used for translational studies.

High throughput fabrication of 3D culture systems is of high interest to stakeholders like pharmaceutical companies, alternative testing industries, and NGOs and needs much research progress [50, 51]. High throughput testing platforms are required as they produce robust data that sheds more light on disease progression, drug efficacy, toxicity studies, and precision medicine applications. 3D bioprinting can be used to develop high throughput testing platforms that can serve as pre-clinical models that are highly reproducible, use fewer reagents, and produce sufficient data at a high speed. Research into high throughput fabrication of cancer models using 3D bioprinting has been shown [52–54]; however, we focused on balancing microenvironmental cues by using dECM-based hydrogel for our cancer model development as well as high throughput fabrication using extrusion-based 3D bioprinting.

Since we aimed to provide breast cancer cells with a three-dimensional microenvironment with biomimetic ECM components, we worked with previously known extrusion-based capability [20, 41, 48, 49] of DAT hydrogel. Previously mentioned studies had bioprinted DAT hydrogel-based bioink in 3D grids in a simple lattice shape; however, we developed a high throughput bioprinting approach for 3D breast cancer constructs or scaffolds to develop a platform for drug screening application. Usually, dECM-based bioprinting has a limitation, making volumetric 3D bioprinting unachievable as the process requires a highly viscous bioink to attain a mechanically stable printed construct. We could circumvent this tissue engineering challenge as 3D microtumors did not pose a volumetric problem during the high throughput fabrication process, and we could easily use a dECM-based hydrogel for this purpose. We focused on fabricating biomimetic bioprinted 3D scaffolds that could be replicated rapidly in a 96-well plate format in a user-friendly manner. This biofabrication technique gave us a large number of 3D biomimetic tumor scaffolds, which served as high-throughput experimental samples for drug screening. We used two anti-cancer agents in this study. 5-fluorouracil, a pyrimidine antagonist that causes genotoxic stress, has multiple mechanisms that can induce cytotoxicity in cancer cells and is already used in clinical settings to treat TNBC patients [55]. PRIMA-1^{Met} belongs to a class of small agents called p53 reactivators that restore mutant p53 into wild-type-like conformation and lead to transactivation of the p53 function [56–58]. This p53 reactivator has been studied for its anti-cancer effects [59, 60], and a single study showcased results for phase II clinical trials for the treatment of hematologic malignancies and prostate cancer [61].

This study presents a high throughput biofabrication method for 3D bioprinting high-density cancer models using DAT-based bioink as a pathophysiological relevant drug screening tumor model. Using detergent-free decellularization for caprine-origin white adipose tissue for DAT hydrogel to formulate dECM-based bioink without adding any additional biomaterial, we encapsulated MDA-MB-231 triple negative, mutant p53 cells and 3D bioprinted lattice-like structures. We then assessed post-bioprinting viability and gene expression profile for the 3D bioprinted TNBC model compared to cells grown in 2D monolayer conditions. We modified the printing protocol to allow high throughput DAT-based bioink printing in a 96 well-plate format. We further characterized printing parameters like nozzle sizes, printing pressures, and printing speeds. We also compared the resultant bioprinted 3D microtumors to manually pipetted scaffolds, along with volumetric characterization of hydrogel dispensed for each bioprinted 3D microtumor. We performed biological characterizations for 3D microtumors bioprinted using two different nozzle sizes and then performed a drug screening assay to validate the high throughput bioprinted 3D microtumors. We also showcased imaging and gene expression analyses on drug-treated, high-throughput, bioprinted 3D microtumors. We also optimized and characterized DAT-based bioink with a heterogeneous cell population to fabricate a high throughput, immunocompetent 3D bioprinted breast cancer model, which we further biologically characterized to explore tumor heterogeneity in our 3D cancer model. Our study showcased that the high throughput printing approach can be applied to fabricate personalized cancer models. This approach can help develop biomimetic tumor microenvironmental heterogeneity with minimal ECM-based biomaterial within a short time for translational studies to give much more clinically relevant results.

2. Materials and methods

2.1. Synthesis of DAT

We DAT using the non-detergent-based decellularization method as previously optimized and discussed elsewhere [62]. Briefly, caprine omental adipose tissue was obtained from a slaughterhouse. After washing the tissue with deionized distilled water, the tissue was minced into small pieces of about 2–3 mm and subjected to decellularization. We used a non-detergent-free method, a 'Hypo-Hyper' based decellularization, which was performed by washing finely chopped adipose tissue with the salt solutions alternating between hypertonic 1 M NaCl solution and hypotonic 0.1 M NaCl solution with distilled water wash in between for 5 min. After three days of decellularization, tissues were again washed with deionized distilled water to remove any traces of residual salts. Decellularized tissues obtained were then incubated in 100% IPA for 48 h for the delipidation process, along with a batch of native tissue of the same original weight as decellularized tissue for further downstream experiments. This was followed again by washing with deionized distilled water. To prevent contamination of decellularized tissue, they were subjected to an overnight 3% H_2O_2 wash followed by three days of distilled water wash to remove any ions left within the matrix. Finally, the decellularized tissue was subjected to lyophilization and stored at −20 °C until further use.

2.2. Histological assessment

To showcase effective decellularization, we performed histological staining for wet native adipose tissue and DAT. Pre-lyophilized native adipose and decellularized tissue were fixed in 4% formalin, embedded in OCT compound (Thermofisher Scientific, US), and then taken for cryosectioning. Sections of 10-micron thickness were generated and taken up for hematoxylin and eosin staining and further for immunofluorescence staining using Hoechst 333 258 (Invitrogen, UK). Microscopic imaging was done to showcase the efficacy of decellularization.

2.3. Morphological SEM analysis

Lyophilized fibrils of native adipose tissue and DAT samples were subjected to gold surface coating using a thin film deposition system (Kurt J. Lesker CMS-18 HV) for 30 s with a 25 mA electric. Representative images were taken from different sections for each test sample. Using a field emission scanning electron microscope (SEM) (JOEL JSM-7800 F Japan) at two different magnifications in vacuum conditions, SEM images were taken while operating under an accelerated voltage of 15 kV.

2.4. DAT hydrogel preparation

For our study, we prepared 3% w/v DAT hydrogel, which was prepared by weighing a sufficient amount of lyophilized tissue and solubilizing it in a solution containing 0.5 M acetic acid with 10% (w/w) pepsin with respect to the weight of the tissue. After solubilization for 24 h, the pH of the pregel was adjusted to 7.4 by adding cold 10 M of NaOH. During the process of pH adjustment, the pregel was always kept at a cold temperature to avoid gelation. pH-adjusted hydrogel was stored in a refrigerator at 4 °C, and each batch was used for experimentation within a week of preparation. Thermal crosslinking or gelation at 37 °C was used to form stable 3D hydrogel scaffolds from the hydrogel. The crosslinking time was optimized for pH-adjusted DAT hydrogels by checking their stability in PBS and was kept uniformly for 30 min of thermal crosslinking for all the experiments in the study.

2.5. Swelling property

A swelling property experiment was performed to assess the swelling behavior of DAT hydrogel in an aqueous environment, as described elsewhere [63]. Briefly, 200 μl of DAT hydrogel was incubated for thermal crosslinking and was weighed and immersed in PBS buffer at 37 °C. At different experimental time points, the 3D hydrogels were taken out of the solution, and the hydrogel was patted with wet filter paper and subsequently weighed. The degree to which the hydrogel swelled in response to PBS was measured as the swelling index and calculated using the formula % swelling $= [(W_\mathrm{f} - W_\mathrm{i})]/W_\mathrm{i}\ {*}100$ where W_f is the weight of hydrogel at experimental time points day 1, 4 and 7 and W_i is the initial gel weight taken on the day of crosslinking.

2.6. Rheological assessment

pH-adjusted DAT hydrogel was subjected to gelation kinetics using the rheometer Kinexus Pro+ (Malvern Panalyticals). A temperature sweep was performed with the ramp rate of 0.5 °C per minute at a gap of 0.3 mm with shear strain fixed at 0.1% and a frequency of 1 Hz to understand the gelation kinetics of 3% DAT hydrogel. This was followed by a time sweep to understand the change in modulus of the hydrogel at a crosslinking temperature of 37 °C, with the measurements taken for an incubation time of 60 min. We performed a shear rate sweep from 0.01 to 1000 s^{-1} at 20 °C with a 0.3 mm gap and frequency fixed at 1 Hz to assess the DAT hydrogel flow property.

2.7. Cell culture

MDA-MB-231 breast cancer cells (NCCS, Pune) were cultured in Minimal eagle medium (Lonza, Europe), NIH-3T3 embryonic mouse fibroblast cells (NCCS, Pune) were cultured in DMEM (Himedia, India), and THP-1 monocytes (NCCS, Pune) were cultured in RPMI (Himedia, India) supplemented with 10% fetal bovine serum (FBS) (Biowest, France) in 37 °C, 5% CO_2 humidified environment. Confluent cells were detached from the 2D tissue culture plates (TCPs) with 0.25% trypsin solution, centrifuged at 1800 rpm for 5 min, suspended in 1 mL fresh 1 X MEM media, and counted.

2.8. Bioink preparation

We optimized bioink extrudability and bioprinting using a 3D printer from Next Big Innovation Labs® with an analog pressure dial. Components like printer nozzles, syringe barrels, and plungers were sterilized with 70% ethanol and UV exposure for 30 min before printing, followed by thorough washing with autoclaved distilled water before loading bioink for bioprinting.

Bioink preparation was done over ice to prevent crosslinking of pH-adjusted DAT hydrogel. For the bioink preparation, the 1 ml pH-adjusted freshly prepared DAT hydrogel was mixed with 10% of 10X concentrated culture media (Sigma-Aldrich, USA) and centrifuged at high speed to remove air bubbles introduced during mixing. For the bioink preparation with the MDA-MB-231 cell population, breast cancer cells were suspended in DAT hydrogel at a density of 1×10^6 cells ml^{-1} for lattice printing experiments. The cell-encapsulated DAT bioink was filled into the barrel, and the air bubbles were removed again from the barrel using centrifugation.

2.9. 3D bioprinting

2.9.1. High throughput 3D bioprinting for drug screening

We used a modified 3D printing approach to attain a 96-well plate format high-throughput 3D bioprinting protocol. A barrel fitted with a general-purpose dispense stainless steel tip with a nozzle with an internal diameter of 330 μm was filled with MDA-MB-231 breast cancer cells laden 3% HH DAT hydrogel (seeding density 3 million cells ml^{-1} hydrogel) and loaded into the cold extruder tool head of NBIL Printer. The printing was conducted at a temperature of 15 °C at a speed of 800 mm min^{-1}, and the printing pressure (operated manually) was set at 0.3 bars after process optimization. Both acellular and cellular inks were tested for process optimization, and the bioprinted 3D microtumors were assessed for biocompatibility. In one printing session, four 96-well plates were printed at most to maintain sterility during the bioprinting process, and the holding time or printing session was never over an hour to ensure cell viability through the process.

All printed constructs were incubated for 30 min at 37 °C for thermal crosslinking, and cell culture media was added subsequently. Printed 3D microtumors were characterized using a bright field microscope to assess morphology as well as cell growth over the culture period, and cell viability for bioprinted constructs was assessed using the Live/Dead assay method described in section 2.10, and Z stacking for 3D constructs was done as described in section 2.12.

2.9.2. 3D bioprinting of heterogeneous breast cancer model

To optimize the printing of multicell-laden bioink, we tested a lattice geometry of size 15 × 15 × 1 mm^3. The barrel fitted with a Nordson general-purpose dispense stainless steel tip with a 22 G nozzle (needle with an internal diameter of 410 μm) was filled with cell-laden 3% HH DAT hydrogel-based bioink and loaded into the cold extruder tool head with the printing temperature set to 15 °C. For the bioink preparation of heterogeneous cancer models, MDA-MB-231 breast cancer cells, NIH-3T3 fibroblasts, and THP1 monocytes were suspended in DAT hydrogel at a density of 2 × 10^6cells ml^{-1} each in a ratio of 1:1:1 with a final seeding density of 6 million cells ml^{-1}. The cellular bioink was printed layer-by-layer, and the bilayered construct was assessed for printing process optimization. Printing pressure was set at 0.3 bars while different printing speeds of 400, 600, and 1000 mm min^{-1} were tested for optimization. High throughput bioprinting using the multicellular bioink was done following printing process optimization. 3D printed constructs were characterized using a bright field microscope to check for strand fidelity, and cell viability for high throughput bioprinted constructs was assessed using the Live/Dead assay method described in section 2.10 as well as confocal imaging for different cellular markers was done as described in section 2.12.

2.10. Cell viability and metabolic activity assessment

Calcein-AM (Invitrogen, UK) and Propidium iodide probe (Sigma-Aldrich, Germany) were used to perform Live/Dead assay to identify live cells (stained green) and dead cells (stained red), respectively, based on the cell membrane integrity. Live/dead was performed on the sample at different time points depending on experiments to evaluate the viability of cells within 3D bioprinted constructs. To check the cell metabolic activity at the endpoint of drug screening experiments, MTT reagent (Sigma Aldrich) was added to the drug-treated sample, with the untreated sample as positive control and the DMSO-treated sample as vehicle control. MTT was incubated for 3 h, followed by DMSO incubation for another 30 min to solubilize formazan crystals. The printed scaffolds were mechanically crushed to ensure all the formazan crystals came out in the DMSO solution and were solubilized. This was followed by measurements of absorbance using a plate reader at a wavelength of 570 nm and 630 nm.

2.11. Drug screening

The high throughput bioprinted platform was validated by exposing the 3D bioprinted microtumors to drug treatment using two anti-cancer drugs-5-fluorouracil (Sigma Aldrich) and a p53 reactivator called PRIMA-1^{Met} (Sigma Aldrich). A stock solution was prepared in DMSO (Sigma Aldrich) with 1 mg ml^{-1} for 5-fluorouracil preparation and 1.25 mg ml^{-1} PRIMA-1^{Met} preparation as per the manufacturer's instructions, and all the working drug concentrations were made by diluting the stock solution in MEM. A combination of drugs was formulated and termed as F5P40, F50P25, and F250P10, referring to 5-flurouracil and PRIMA-1^{Met} with digits following referring to respective concentrations in micromolar. The 3D bioprinted microtumors were exposed to varying concentrations as a single dose, and a combination of both drugs and untreated experimental control and DMSO-treated vehicle control were kept. The duration of drug treatment was 72 h, and the drug cytotoxicity was evaluated within the 3D microtumors using an MTT assay to assess metabolic activity as described in section 2.10 and plotted as bar graphs. Cellular viability was assessed for drug-treated microtumors through Live/Dead Assay using Calcien-AM to stain live cells green and PI to stain dead cells, as described in section 2.10, and representative images from experimental replicates have been shown. Samples were also taken for total RNA extraction using the standard Trizol method.

2.12. Immunohistochemistry and immunofluorescence analysis

Lyophilized DAT was fixed using 4% paraformaldehyde and then stained with hematoxylin (Sigma) and eosin (Sigma) as well as Hoechst 33 258 (Invitrogen, UK) to assess the efficiency of decellularization. The thickness of DAT hydrogel-based high throughput 3D bioprinted microtumors using different nozzles was assessed. Bioprinted microtumors were fixed after two days of culture in 4% paraformaldehyde and stained with Hoechst 33 258 (Invitrogen, UK). Heterogeneous 3D bioprinted breast cancer models were fixed after day one and day seven of culture for confocal imaging to assess the cellular population and functionality. Constructs were fixed at time points of day 1 and 7 in 4% formaldehyde and stained with the markers E-cadherin (4 A2) Mouse mAb #14472 (Cell Signaling Technology, USA), mutant p53 monoclonal antibody (Invitrogen, UK), PDGFRα monoclonal antibody (Thermofisher Scientific), Vimentin (D21H3) XP® Rabbit mAb (Cell Signaling Technology, USA), CD14 (Cell Signaling Technology, USA), CD68 monoclonal antibody (Invitrogen, UK), and CD44 polyclonal antibody (Invitrogen, UK), as well as Hoechst 33 258 (Invitrogen, UK), was used immunofluorescence experiments. Primary antibodies were used in a concentration of 1:500, while secondary antibodies, anti-rabbit Alexa Fluor 594 (Cell Signaling Technology, USA) and anti-mouse Alexa Fluor Plus 647 (Invitrogen, UK), were used in a concentration of 1:200.

2.13. Gene expression analysis

Total RNA from cells was isolated using the standard Trizol method from breast cancer cells grown in 2D TCP and 3D bioprinted cancer models at different time points according to experiments using Ambion TRIzol™ Reagent from Thermo Fisher Scientific. For high throughput 3D bioprinted constructs, due to their smaller size, sufficient constructs were pooled together for each drug-treated group to obtain optimal RNA concentration. The obtained total RNA was dissolved in DEPC (Diethylpyrocarbonate) treated water, and the A260/A280 ratio was calculated using the NanoDrop™ 2000 Spectrophotometer. Only samples with an A260/A280 ratio of 1.8–2.0 were included for cDNA preparation. From all the test samples, volumes with a constant RNA concentration of 25 ng μl^{-1} were reverse transcribed to cDNA (Takara Bio's PrimeScript 1st strand cDNA Synthesis Kit). For the final reaction mixture, 25 μl of reaction mix was made using Takara Bio's TB Green Premix Ex Taq II (Tli RNase H Plus), wherein both forward and reverse primer had a final concentration of 1 pmol μl^{-1} in the total reaction mix. Forward and reverse primer sequences have been included in supplementary information. We used Qiagen Rotor Gen Q PCR Machine, and the PCR cycles had a denaturation phase set at 95 °C for 20 s, followed by an annealing phase at 95 °C for 20 s and an extension phase at 72 °C for 30 s with a total of 40 reaction cycles per run. At the end of each reaction, ΔCT values for each gene of interest were calculated in triplicates and normalized to ΔCT values for the housekeeping gene GAPDH. The $\Delta\Delta CT$ method was used for gene expression analysis, and relative fold change was obtained and analyzed.

2.14. Statistical analysis

Unless mentioned, all the experiments were conducted in triplicates, and data are represented as mean ± sd. All statistical analysis was performed on GraphPad Prism software, and significance was checked using the student t-test and ANOVA. $^{***}p < 0.001$, $^{**}p < 0.01$, $^{*}p < 0.05$ were considered as statistically significant.

3. Results

3.1. Decellularization of adipose tissue and preparation of dECM-based hydrogel

We performed decellularization of white adipose tissue procured from a slaughterhouse based on a non-detergent-based protocol. Herein, we used salt-based solutions with 0.1 M NaCl hypotonic and 1 M NaCl hypertonic solutions sequentially to remove cellular content within adipose tissue and 100% IPA washes to delipidate the adipose tissue. Figure 1(A) summarizes all the steps of decellularization. The efficacy of the decellularization protocol was ensured by histological sections, as shown in figures 1(B) and (C), comparing native and DAT. A lack of nuclear content and cellular debris was seen, while preserved ECM proteins can be seen in the hematoxylin and eosin-stained DAT (lower panel) section compared to native adipose tissue (upper panel), where cells with nuclear content are evident. Further, SEM imaging from lyophilized tissues showed the morphology of retained ECM fibrils in DAT.

We performed a gelation kinetics experiment to understand the thermal crosslinking of DAT hydrogel. The DAT hydrogel gradually showed an increase in complex modulus from 70 Pa in the initial 5 min at 4 °C to 1570 Pa at the end of the 1-hour incubation period at 37 °C, indicating sol-to-gel transition as seen in figure 1(D). Figure 1(E) shows thermally crosslinked 3% w/v DAT hydrogels incubated in PBS to showcase the stability of crosslinked hydrogel. The swelling index demonstrated in figure 1(F) showed no significant difference between different time intervals from day 1 to day 4 and day 4 to day 7. However, a slight increase in the swelling index was seen from 11.524% ± 4.18% on day 1–21.598% ± 4.78% on day 7.

3.2. 3D bioprinting of breast cancer model using DAT hydrogel

To ensure 3% w/v DAT hydrogel was printable, we did a shear rate sweep that showcased that viscosity decreased with an increase in shear rate, as seen in figure 2(A). This indicated the shear thinning property of the hydrogel and indicated that the hydrogel would be suitable for extrusion-based 3D bioprinting. Next, we prepared a bioink of MDA-MB-231 breast cancer cells encapsulated in DAT hydrogel and used extrusion-based bioprinting to fabricate a 3D cancer model. In figure 2(B), we have compared the morphology and cellular proliferation of the cells on conventional 2D TCP and 3D bioprinted cancer models. Breast cancer cells showcased their stellate morphology post 3D bioprinting. The results showed the viable cell population post-printing and proliferation over seven days of culture, showcasing the cytocompatibility of DAT hydrogel and its use for 3D printing applications.

To further confirm the cellular functionality post-3D bioprinting process, a gene expression profiling was done for a set of genes using qRT-PCR. As results show in figure 2(C), a significant increase in gene expression for cancer stemness, namely Oct4 (octamer binding transcription factor 4) and Sox2 (sex determining region Y—related high mobility group box 2), as well as gene expression for MMP2

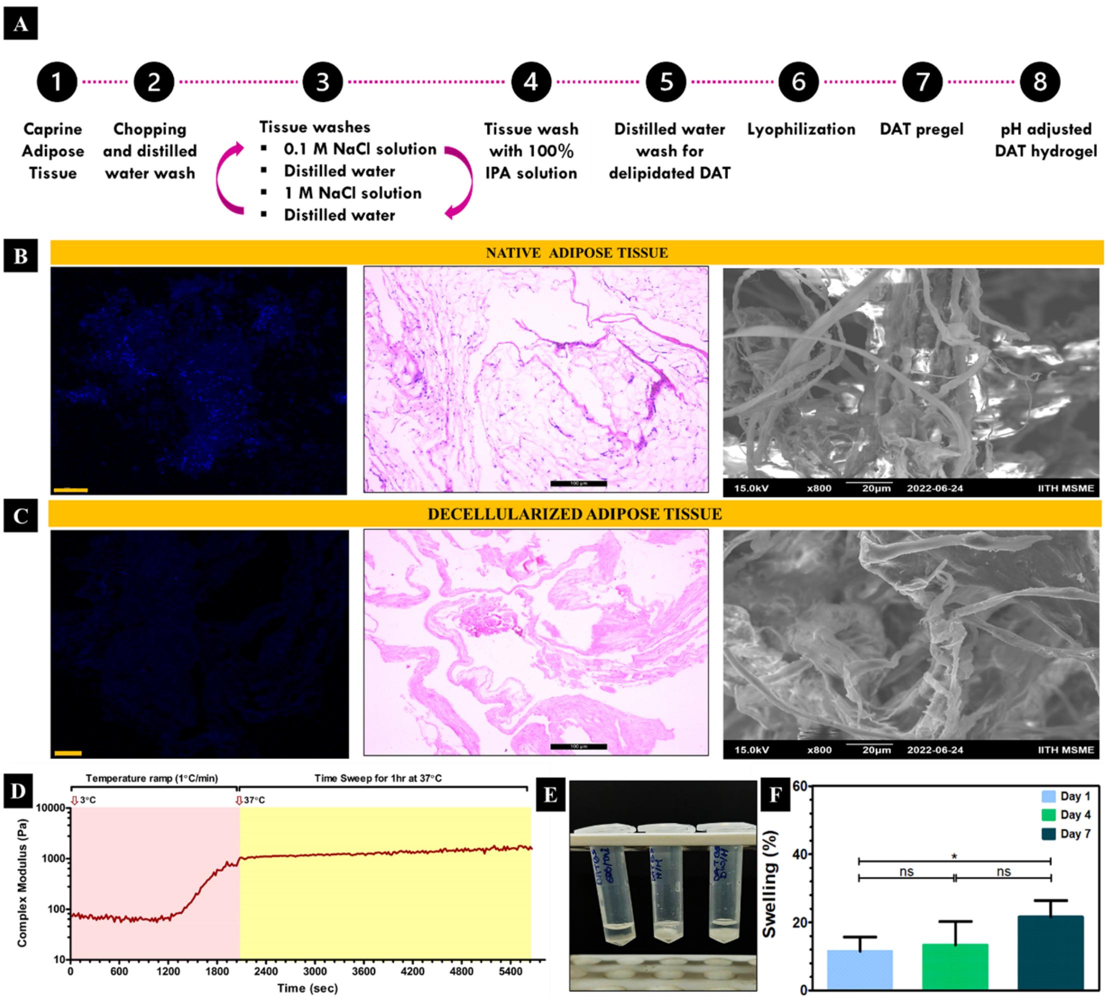

Figure 1. Decellularization of adipose tissue (DAT): (A) A summary of steps for decellularization used in the study (B) and (C) Hoechst 33 258 staining, HE staining, and SEM images of native adipose tissue (upper panel, scale bar = 100 μm) and decellularized adipose tissue (lower panel, scale bar = 200 μm) (D) Gelation kinetics of pH adjusted digested DAT hydrogel (E) Stability of thermally crosslinked DAT hydrogel in PBS post crosslinking (F) Swelling property of DAT hydrogel over seven days of incubation in PBS. ($n = 3$, $^{*}p < 0.05$, $^{**}p < 0.01$, and $^{***}p < 0.001$).

and MMP9 (Matrix metalloproteinase 2 & 9) correlating to cancer cell invasion and migration, was observed in 3D bioprinted samples as compared to cancer cells grown on 2D TCP.

3.3. Process optimization of high throughput printing protocol

Once the feasibility of using DAT hydrogel as bioink was seen and the cytocompatibility of the bioprinting process was affirmed, we modified the 3D bioprinting protocol to fabricate 3D scaffold or '3D microtumors' in a high throughput manner within 96 well-plates. Figure 3(A) showcases the bioprinting approach in a 96-well plate, with the zoomed-in image showcasing a top view of printed scaffolds. The printing process for a single plate took 7 min compared to manual pipetting, which took 15 min. After the printability was assessed to be possible using dECM hydrogel, we attempted to characterize the high throughput printing using multiple nozzle sizes, as shown in figure 3(B). 3D Printing was performed across a range of pressures to see the print fidelity of the scaffold and assess the diameter of the 3D bioprinted construct or microtumor. When we changed the pressure from 0.2 bars to 0.7 bars and the printing nozzle size, we observed significant differences between the diameters of bioprinted constructs using different nozzle sizes. For the 150 μm nozzle size, the range of diameter of microtumor was 620.63 $\pm$ 28.04 μm to 846.96 $\pm$ 32.65 μm when the pressure was changed from 0.2 bars to 0.7 bars. For 250 μm nozzle size, the microtumor diameter range was 669.56 $\pm$ 17.64 μm to747.02 $\pm$ 57.23 μm when the pressure was changed from 0.2 bars to 0.7 bars. Similarly, for 330 μm nozzle size, the range of diameter of microtumor was 546.84 $\pm$ 22.73 μm to 824.41 $\pm$ 21.84 μm when the pressure was changed from 0.2 bars to 0.7 bars. Irrespective of nozzle sizes, the diameter of 3D bioprinted scaffolds can be influenced by extrusion pressure.

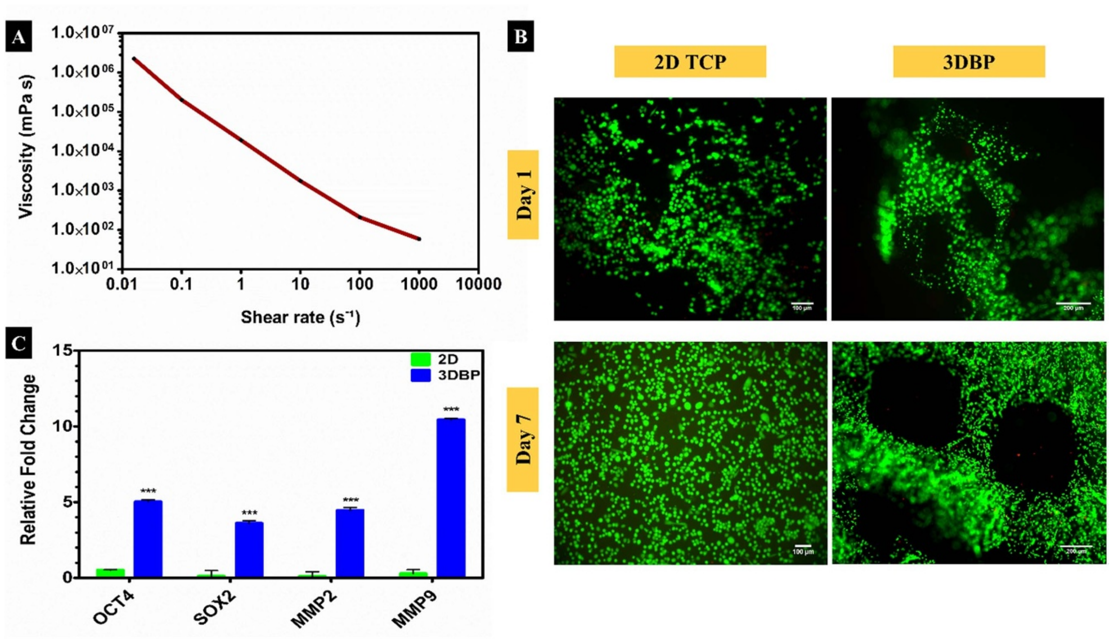

Figure 2. DAT hydrogel-based 3D bioprinted breast cancer model: (A) Rheological profile of DAT hydrogel with shear rate sweep at 20 °C (B) Cellular morphology and proliferation of MDA-MB-231 breast cancer cell in 2D tissue culture plate (TCP) and DAT hydrogel-based 3D bioprinted cancer model on day one and day seven (live/dead assay image representing viable cells in green and dead cells in red). (C) Gene expression analysis for cells cultured till day seven on 2D TCP and encapsulated in a 3D bioprinted cancer model. ($n = 3$, $^{*}p < 0.05$, $^{**}p < 0.01$, and $^{***}p < 0.001$).

A previous study has shown that nozzles with inner diameters larger than 200 μm should be used for better printing accuracy and replicability [64]. Therefore, we chose to characterize further scaffolds printed with 250 μm and 330 μm for further experiments as we observed more deposition of DAT hydrogel with 150 μm nozzle indicated by their respective printed scaffolds with larger diameters.

We checked the diameters of scaffolds printed with nozzle sizes of 250 μms and 330 μms in comparison to the 3D DAT scaffolds deposited with conventional manual pipette aid, and quantified results are shown in figure 3(C). It was observed that there was a non-significant difference in microtumor diameter when bioprinted using a 250- and 330-micron nozzle at 0.3 bars of pressure. There was a significant difference between the diameter of scaffolds deposited with a 250-micron nozzle, which was 634.85 $\pm$ 18.47 μm compared to 2 μl and 5 μl DAT droplet dispensed manual pipette aid. Similarly, a 3D bioprinted microtumor fabricated at 0.3 bars of pressure using a 330 μm nozzle had a diameter of 713.71 $\pm$ 22.20 μm. It showed a significant difference compared to the diameter of 2 μl and 5 μl liquid droplets dispensed from manual pipette, which were 983.0844 $\pm$ 12.92 μm and 1383.496 $\pm$ 59.49 μm respectively. The viscosity of the hydrogel can directly influence the deposition of material in response to extrusion pressure and, therefore, impact the diameter of the 3D bioprinted scaffold.

The next parameter we assessed was the thickness of the 3D bioprinted scaffold or the depth of hydrogel deposited in the form of a 3D bioprinted microtumor. The data was collected using Z stacking during confocal imaging of bioprinted constructs using nozzle sizes 250-micron and 330-micron. Figure 3(D) showcases that the 3D microtumors printed using a 330-micron nozzle had more depth or higher scaffold height of 476.666 $\pm$ 56.27 μm, whereas the 3D microtumors printed with a 250-micron nozzle showed a significantly lower depth or lower scaffold height of 120.476 $\pm$ 46.20 μm in comparison.

3.4. Characterization of high throughput bioprinted cancer model

3.4.1. Volumetric characterization of high throughput 3D bioprinted microtumors

We wanted to quantify the amount of DAT hydrogel being deposited in each well when using the 250- and 330-micron nozzle, and we used a method discussed elsewhere [64]. Table 1 summarizes the relevant weights taken for printed scaffolds after multiple bioprinting rounds, and we calculated the final average volume of 3.47 μl and 4.51 μl hydrogel dispensed from 250 and 330 μm nozzle for each bioprinted scaffold in each well based on the density of hydrogel being 1 kg m^{-3}. The rationale behind this experiment was to gain an insight into the volume of bioink being extruded out to print on a 96-well plate with error-free deposition of 96 scaffolds while using 250- and 330-micron nozzle.

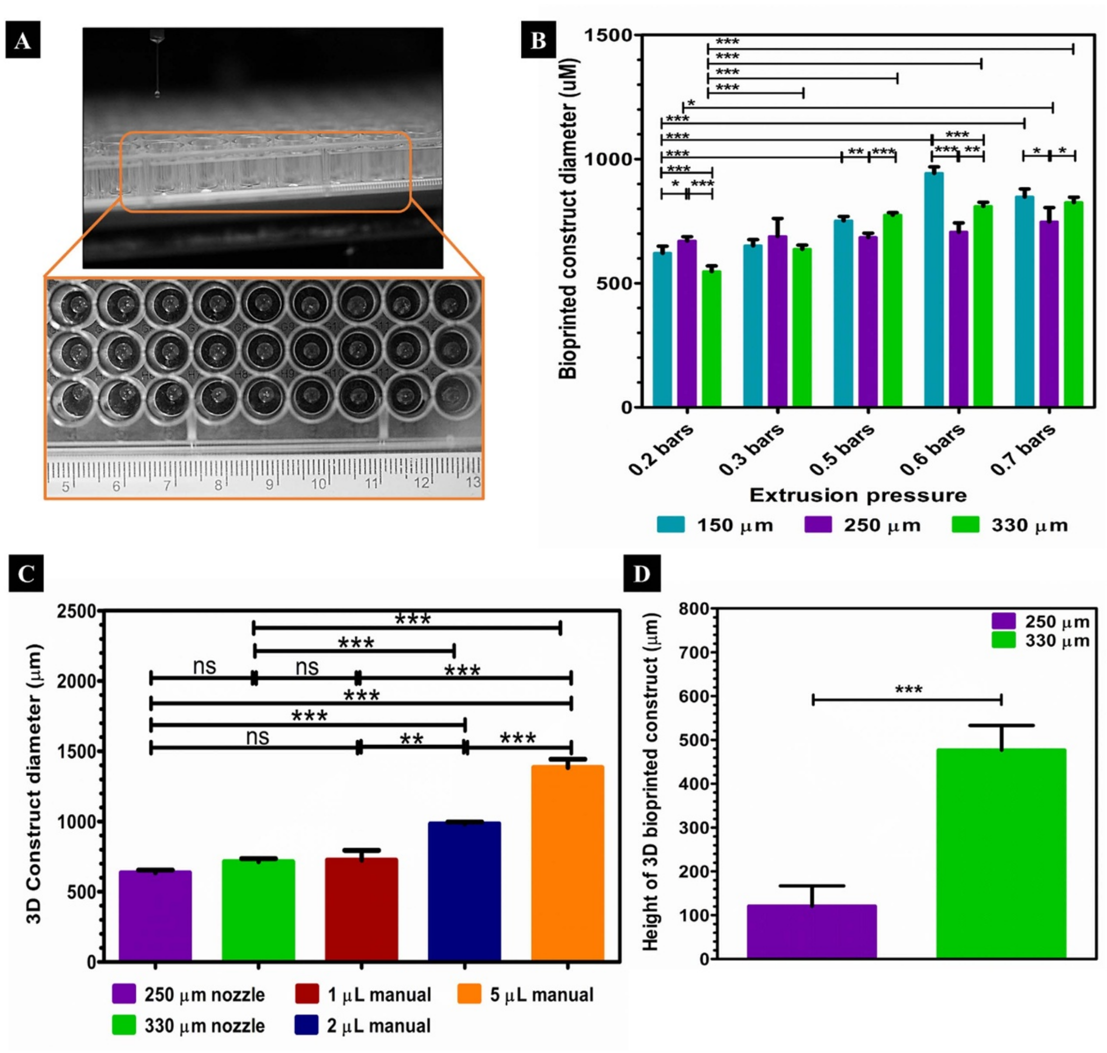

Figure 3. Process optimization for DAT hydrogel-based high throughput 3D printing. (A) High throughput 3D printing protocol adapted to a 96 well-plate format (inset image shows the dispersion of bioink within the wells) (B) Optimization of 3D bioprinted constructs diameter using nozzles of different diameter at varying printing pressures. (C) Comparison of the diameter of 3D bioprinted constructs using different nozzles with manual pipette aid. (D) Quantification of Z stacks for comparison of height of 3D bioprinted constructs using different nozzles. ($n = 3$, $^*p < 0.05$, $^{**}p < 0.01$, and $^{***}p < 0.001$).

Table 1. Volumetric characterization of 3D bioprinted constructs extruded using high throughput printing protocol.

Size of nozzle (Inner Diameter)	Weight of empty 96-well plates ($n = 3$)	Weight of printed 96-wellplates ($n = 3$)	Weight of all printed scaffolds	Weight of single printed scaffold	Density of hydrogel	Volume of single printed scaffold
250 μm	61 392.1 mg	61 393.1 mg (288 scaffolds)	1 mg	0.00347 mg	1 kg m^{-3}	3.47 μl
330 μm	61 622.1 mg	61 623.4 mg (288 scaffolds)	1.3 mg	0.00451 mg	1 kg m^{-3}	4.51 μl

3.4.2. Biological characterization of high throughput 3D bioprinted microtumors

Next, we wanted to assess the biocompatibility of the high throughput bioprinting process. Therefore, we characterized the biocompatibility of the 3D bioprinting process using two nozzle sizes- 250-micron and 330-micron. Figures 4(A)–(D) showcases biological characterization for high throughput printed 3D microtumors using a 250-micron nozzle. Similarly, figures 4(E)–(H) showcases biological characterization for high throughput printed 3D microtumors using the 330-micron nozzle. Figures 4(A) and (I) depict bright-field microscopic images of bioprinted 3D microtumors using 250- and 330-micron nozzle, showcasing cell-laden constructs with spherical and spindle-shaped encapsulated MDA-MB-231 cells. Figures 4(B) and (F) showcase viable cells stained in green and dead cells stained in red

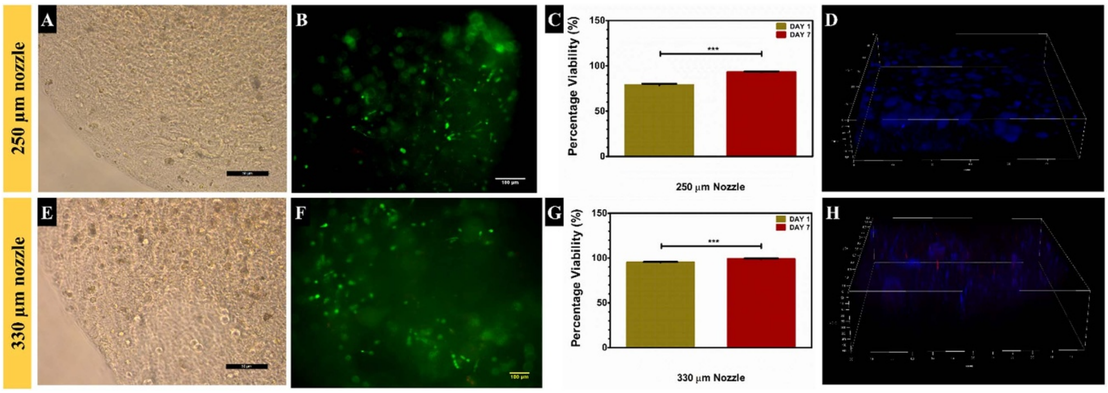

Figure 4. High throughput 3D bioprinting of MDA-MB-231-laden DAT bioink in a 96-well plate using (A)–(D) 250-micron nozzle and (E)–(H) 330-micron nozzle. (A), (E) Morphology of encapsulated cells within bioprinted construct at day 1 post-printing (scale bar: 50 μms); (B), (F) cell viability within 3D microtumor at day 7 post-printing (scale bar: 100 μms); (C), (G) quantification of cellular viability from live/dead assay; (D), (H) Z-stack of 3D microtumor at day 7 post-printing (scale bar: 50 μms). ($n = 3$, $^{*}p < 0.05$, $^{**}p < 0.01$, and $^{***}p < 0.001$).

within 3D bioprinted microtumors, showing that the 3D bioprinting using both nozzle sizes supported cellular viability. Quantified data from Live/dead assay, as seen in figures 4(C) and (G), showed a significant increase in cellular viability within both groups of bioprinted scaffolds from day 1 to day 7 of culture.

For all time points, the cellular viability within bioprinted microtumors was high for both nozzle sizes. On day 7, the microtumors bioprinted using a 250-micron nozzle showed a cellular viability of 93.20% $\pm$ 0.61%, compared to a statistically significantly higher cellular viability of 98.84% $\pm$ 0.68% within the microtumors bioprinted using 330-microns nozzle.

Figures 4(D) and (H) showcase representative Z-stack immunofluorescent images of cell-laden 3D microtumor bioprinted using both nozzle sizes and show the cells populating in three-dimensional space.

3.5. Drug screening application of high throughput bioprinted cancer model

3.5.1. Drug screening on high throughput 3D bioprinted microtumors

Biofabrication of a 3D bioprinted cancer model using dECM hydrogel in a high throughput manner was done using the 330-micron nozzle to produce a relevant drug screening platform. Figure 5(A) showcases the experimental protocol and chronology for the anti-cancer drug screening application on the high throughput 3D bioprinted cancer model. Cells were allowed to grow in a 3D bioprinted microtumor for three days, after which they were subjected to drug screening. For this purpose, we selected two anti-cancer drugs and screened for multiple concentrations. 5-fluorouracil (5-FU) is a well-known chemotherapeutic drug that induces genotoxic stress to cause cell death, and PRIMA-1^{Met} is a p53 reactivator being tested in phase 1 clinical trial that causes reactivation and restoration of mutant p53 into wild-type p53 and induces cell death in cancer cells. The rationale behind our selection of drug for our drug response assay was that 5-FU has been used in clinical treatments for TNBC patients and extensively studied, while PRIMA-1^{Met} remains unexplored in clinical settings and has not yet cleared clinical trials for breast cancer. Most clinical cases of TNBC present with a mutation in their p53 gene, making it necessary to explore mutant p53-targeted therapeutics [1–4]. In this study, we chose a combination of both these drugs to assess for any marked change in cytotoxicity of MDA-MB-231 cells, which is the human TNBC cell line. Additionally, we opted to screen for a wide range of concentrations for both drugs, as mentioned in each figure, as we tested these drugs on our novel high throughput bioprinted TNBC model for the first time.

We first used the MTT assay to assess the metabolic activity of cancer cells treated with PRIMA-1 ^{Met} in figure 5(B) and 5-FU in figure 5(C). Treatment with PRIMA-1^{Met} showcased a potent decrease in cell metabolic activity after 50 μM, while treatment with 5-FU showcased a gradual decrease in cell metabolic activity after 500 μM. Overall, a dose-dependent decrease in metabolic activity was observed. Next, we wanted to assess if high throughput 3D microtumors could be imaged, allowing for more downstream analysis and gene expression studies for relevant results.

We further assessed the cytotoxic effect of combinatorial drug doses as shown in figure 5(D), where it was seen that with increasing concentration of PRIMA-1 ^{Met} in the combinatorial dose, the metabolic activity of cancer cells was reduced compared to the untreated control group, showing a synergistic effect. Consequently, the biofabrication of a high number of 3D bioprinted microtumors allowed for a better drug screening setup and allowed us to test a wide range of concentrations.

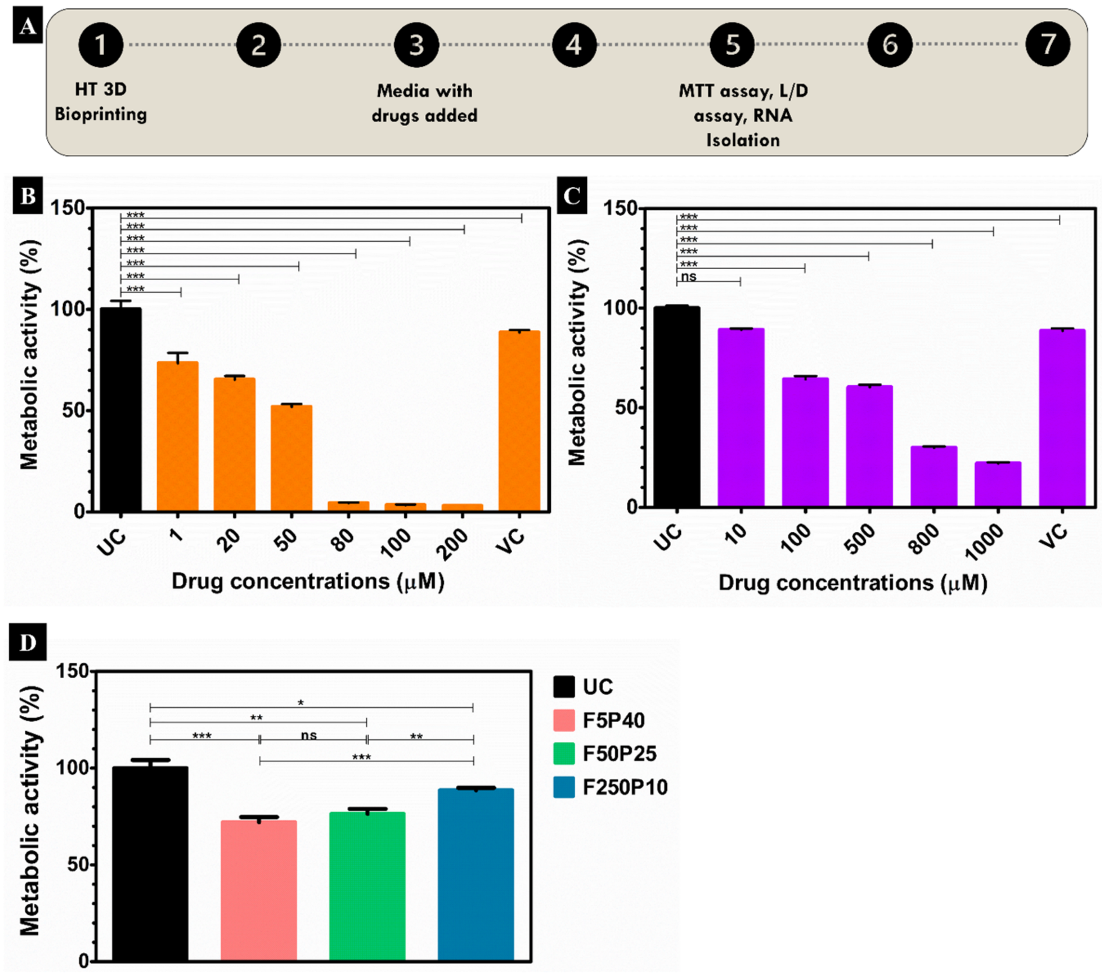

Figure 5. Drug screening application on 3D bioprinted cancer model as a high throughput platform (A) anti-cancer drug screening experimental protocol and chronology. (B)–(D) Metabolic activity of cancer cells after drug treatment using (B) PRIMA-1 ^{Met}, a p53 reactivator (C) 5-Florouracil, a DNA damaging anti-cancer agent, and (D) combinatorial treatment on high throughput 3D bioprinted constructs. ($n = 3$, $^{*}p < 0.05$, $^{**}p < 0.01$, and $^{***}p < 0.001$).

3.5.2. Cellular viability in drug-treated 3D bioprinted microtumors

The high throughput 3D bioprinted microtumors allowed us to perform an imaging analysis of drug-treated samples due to the increased number and small morphology. We treated the 3D microtumors with two different drugs and subjected the untreated and treated 3D microtumors to live/dead staining at the endpoint to detect the viable cell population within the 3D bioprinted microtumors.

Figures 6(A) and (B) showcase the imaging analysis for microtumors treated with PRIMA-1^{Met}, while Figures 6(C) and (D) showcase the imaging analysis for microtumors treated with 5-FU. Figures 6(A) and (C) showcase the representative images from live/dead assay, with the number of dead cells stained red gradually increasing as the drug concentrations increased compared to the untreated 3D bioprinted microtumors. Furthermore, we observed that the number of stressed, dying population of cells increased within 3D bioprinted microtumors exposed to higher concentrations of drugs. The data were quantified in figures 6(B) and (C), where a dose-dependent cytotoxic effect of the anti-cancer drugs was seen with a significant decrease in the viability of MDA-MB-231 breast cancer cells in drug-treated microtumors as compared to untreated microtumors. Similarly, the chemoresistance property of breast cancer cells was observed within these 3D microtumors, with viable cells observed at high concentrations of drugs.

3.5.3. Gene expression analysis for drug-treated 3D bioprinted microtumors

Next, we used a large number of samples that could be fabricated from high throughput 3D bioprinting for gene expression analysis for a panel of genes to assess the response of cancer cells toward chemotherapeutic agents. The gene expression levels for Oct4, Sox2, MMP2, MMP9, and p63 in high throughput 3D microtumors were assessed to observe the effect of anti-cancer drugs in the treated cancer model as compared to the untreated cancer model and plotted in figure 7(A).

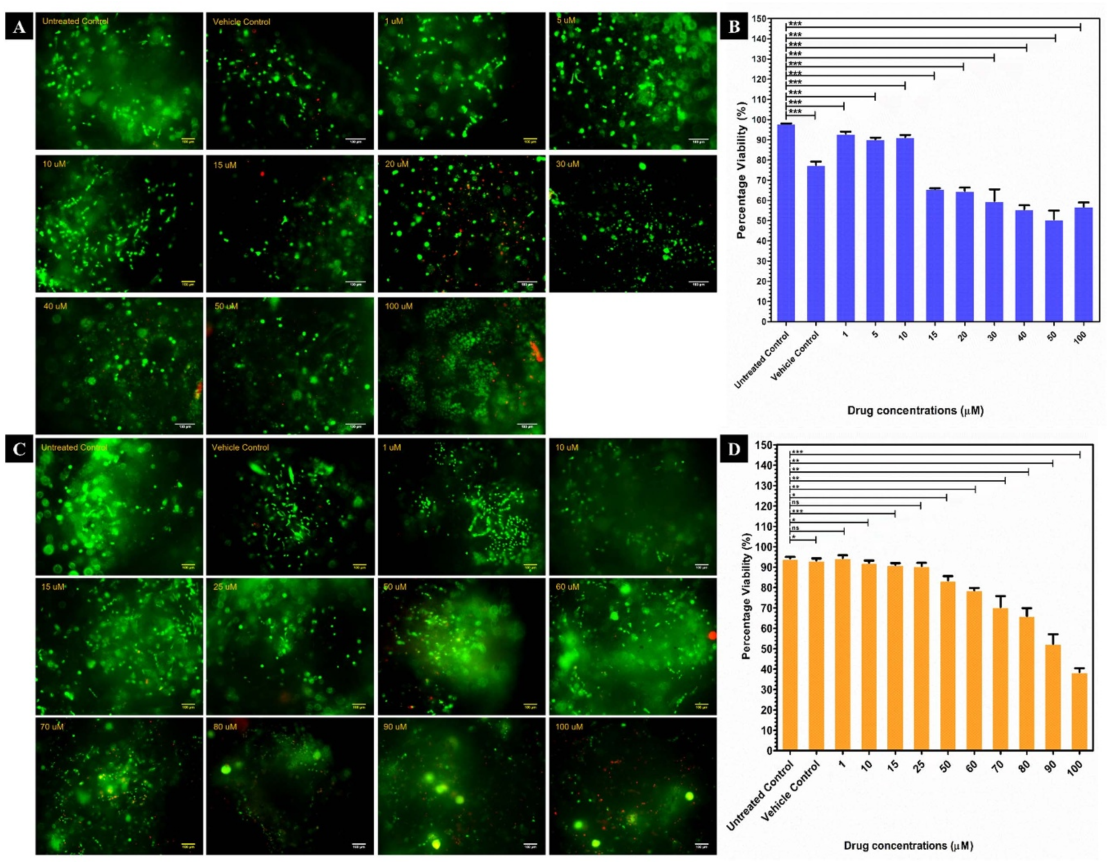

Figure 6. Drug screening on high throughput printed 3D microtumors (A)–(B) Drug treatment on 3D bioprinted microtumors using PRIMA-1 ^{Met} (A) Live/Dead assay showcasing drug-treated 3D bioprinted cancer models (scale bar: 100 μms). (B) Quantification of viable MDA-MB-231 cells within the drug-treated 3D bioprinted constructs. (C)–(D) Drug treatment on 3D bioprinted constructs using 5-fluorouracil (C) live/dead assay showcasing drug-treated 3D bioprinted cancer models (scale bar: 100 μms). (D) Quantification of viable MDA-MB-231 cells within the drug-treated 3D bioprinted constructs. ($n = 3$, $^{*}p < 0.05$, $^{**}p < 0.01$, and $^{***}p < 0.001$).

It was observed that compared to untreated 3D microtumors, the expression levels for the genes Oct4 and Sox2 responsible for cancer stemness, as well as genes MMP2 and MMP9 responsible for cell invasion and migration, were significantly downregulated within treated 3D microtumors with increasing drug concentrations as compared to untreated control. In contrast, p63, a metastatic suppressor gene, was upregulated, although not significantly. The gene expression pattern for the panel of genes was the same with the combinatorial dose, with a downregulation in expression for the genes Oct4, Sox2, MMP2, and MMP9, while p63 was upregulated compared to untreated control.

The gene expression levels for p21 in high throughput 3D microtumors were assessed to observe the effect of anti-cancer drugs and plotted in figure 7(B). It was observed that compared to untreated 3D microtumors, the p21 expression levels were significantly increased within treated 3D microtumors with increasing drug concentrations. Higher p21 expression was noted for combinatorial dose F50P25 compared to F250P10, showing a synergistic effect of drugs on the upregulation of the p21 gene. The increased expression of the p21 gene was observed in samples treated with PRIMA-1^{Met} compared to chemotherapeutic agent 5-FU.

3.6. Biofabrication of immunocompetent high throughput bioprinted 3D breast cancer model

Further, we wanted to check the feasibility of high throughput printing of a heterogeneous breast cancer model incorporating multiple cell populations. We therefore used an extrusion-based 3D bioprinter (Trivima Bioplotter, NBIL, Bangalore, India) to bioprint a multicell bioink composed of MDA-MB-231 cancer cells, NIH-3T3 fibroblasts, and THP-1 monocytes to fabricate an immunocompetent breast cancer model in a high throughput format. The grid or lattice structure was printed to optimize the initial extrusion of heterogeneous cell population-laden bioink and assess the printability of heterogeneous bioink as a preliminary experiment. Lattice geometry was 3D bioprinted to optimize print fidelity, as seen in figure 8(A), with a size of $15 \times 15 \times 1$ mm^3. A layer-by-layer printing approach was done for two layers, each layer of height equivalent to a nozzle diameter of 410 μm, and we have showcased the

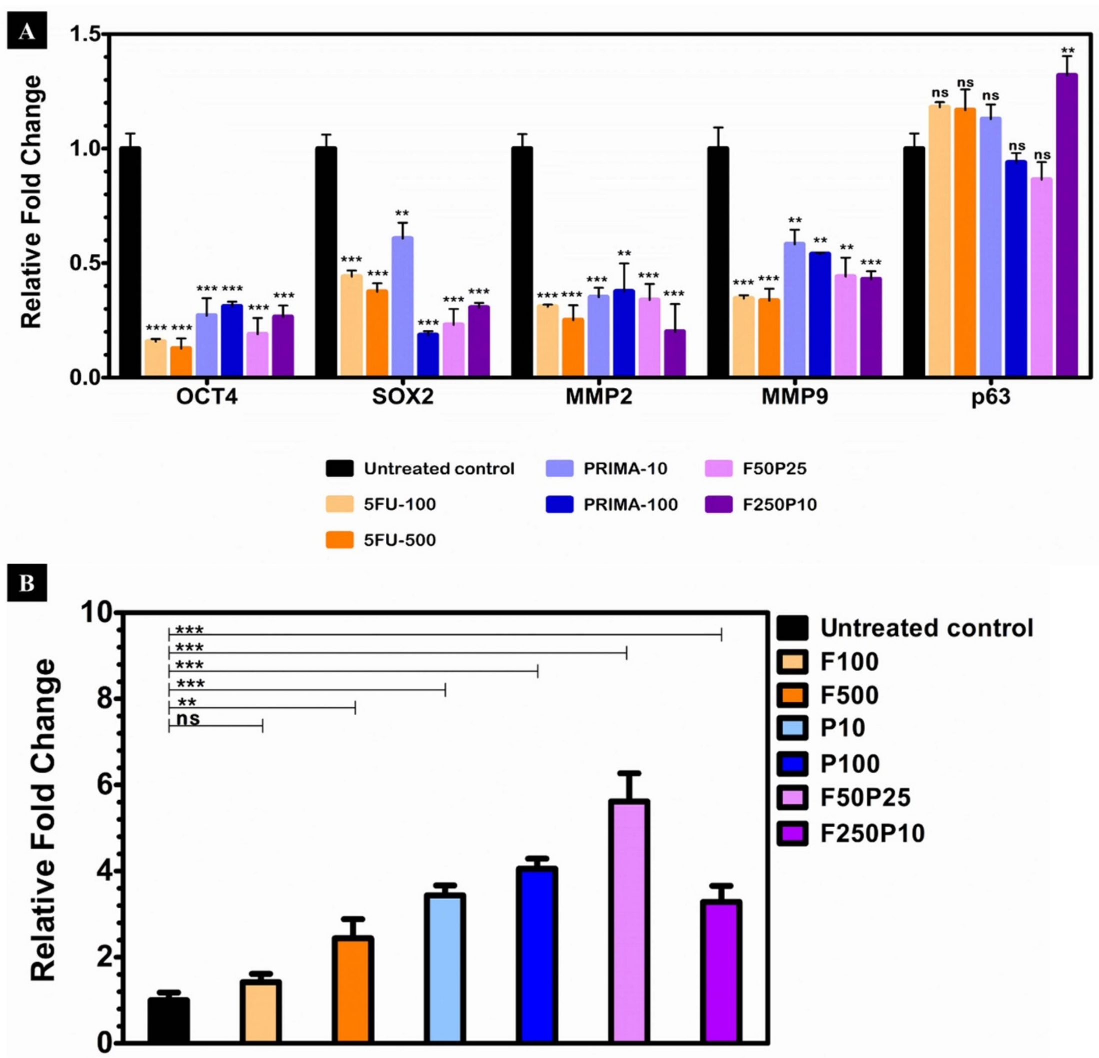

Figure 7. Differential gene expression analysis for untreated and treated high throughput 3D bioprinted microtumors. (A) For genes Oct4, Sox2, MMP2, MMP9, and p63 (B) For the gene p21. ($n = 3$, $^*p < 0.05$, $^{**}p < 0.01$, and $^{***}p < 0.001$).

3D model in supplementary figure 1. Three printing speeds, 400 mm min^{-1}, 600 mm min^{-1}, and 1000 mm min^{-1}, with a similar printing pressure, were used to 3D bioprint the heterogeneous cancer model with DAT hydrogel. A constant pressure of 0.3 bars was applied during printing pressure to ensure the fidelity or resolution of printed strands. We observed that among the constructs fabricated at all three printing speeds, the printing resolution looked better at 1000 mm min^{-1} than at the other speeds. We observed a morphological difference between all three printing speeds, with less spreading of strands in the 3D constructs bioprinted at a higher speed (figure 8(A)). We also printed strands at a higher speed and have showcased the same in supplementary figure 2. Slight spreading occurred at lower speeds of 400 mm min^{-1} and 600 mm min^{-1}; however, it was under the acceptable range, as seen in graphs in figures 8(B)–(D). All the 3D bioprinted constructs were stable during printing and held their layer-on-layer printed strand organization after the bioprinting process.

During lattice bioprinting, we observed significant differences between the expected strand width (410 μm) and the realistic strand width of the bioprinted dECM scaffolds, as quantified in figures 8(B)–(D). The mean width of the vertical printed strand at top layer 2, printed at a speed of 400 mm min^{-1}, was 254.819 $\pm$ 47.12 μm; for strands printed at 600 mm min^{-1}, the width was 298.688 $\pm$ 7.93 μm, and for strands printed at 1000 mm min^{-1} the width was 188.736 $\pm$ 27.29 μm respectively as quantified in figure 8(B). The mean width of the horizontal printed strand in bottom layer 1, printed at a speed of 400 mm min^{-1}, was 323.495 $\pm$ 35.43 μm; for strands printed at 600 mm min^{-1}, the width was 337.729 $\pm$ 3.75 μm, and for strands printed at 1000 mm min^{-1} the width was 271.338 $\pm$ 9.33 μm respectively as quantified in figure 8(C). Figure 8(D) quantifies the bilayered grid

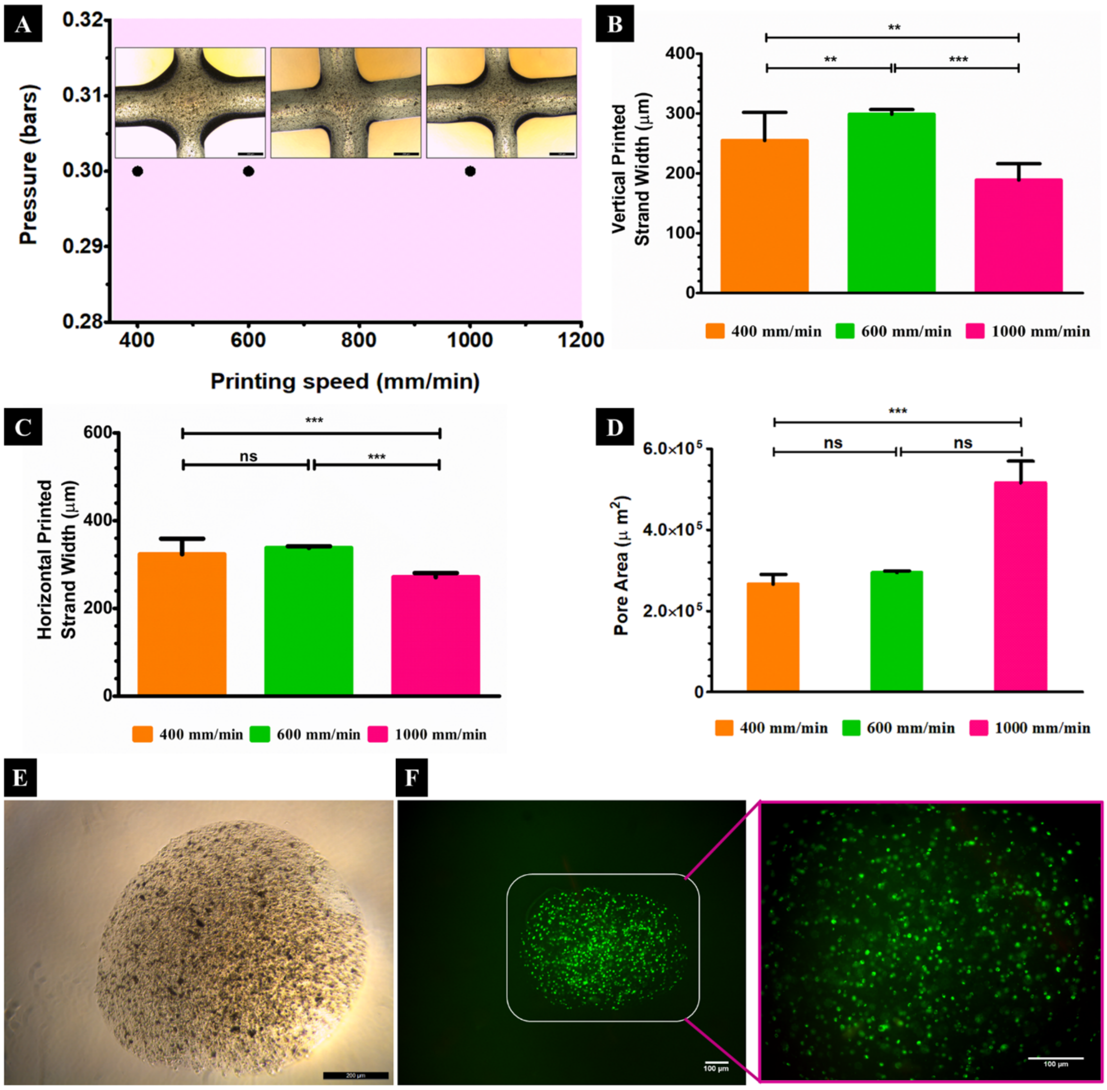

Figure 8. Optimization of extrudability and 3D Bioprinting of multicellular DAT hydrogel. (A) Characterization of immunocompetent TNBC bioprinted model with MDA-MB-231, NIH-3T3, THP-1 cell population fabricated with extrusion-based 3D bioprinting of bilayered lattice structures. Quantification and comparison of (B) 3D bioprinted vertical strand (C) 3D bioprinted horizontal strand width within 3D bioprinted constructs. (D) Quantification and comparison of pore area between horizontal and vertical printed filaments of 3D bioprinted constructs. (E) Microscopic image of a heterogeneous cell populated 3D microtumor (F) Live/Dead assay showcasing viable cell population at day two post-printing within high throughput printed 3D microtumor (inset image shows zoomed closeup). ($n = 3$, $^{*}p < 0.05$, $^{**}p < 0.01$, and $^{***}p < 0.001$).

porosity or pore diameter, which was calculated by measuring the pore area between strands from multiple pore edges.

The mean pore area between printed strands, printed at a speed of 400 mm min^{-1}, was 266 078.89 $\pm$ 23 854.36 μm^2; strands printed at 600 mm min^{-1} was 294 379.417 $\pm$ 4555.32 μm^2, and strands printed at 1000 mm min^{-1} was 516 187.383 $\pm$ 53 274.99 μm^2 respectively. DAT hydrogel-based lattice-shaped bioprinted construct printed at a speed of 1000 mm min^{-1} showcased better retention of pore integrity because of better print fidelity, followed by 600 mm min^{-1}, and the least shape integrity was shown by constructs printed at a lower speed of 400 mm min^{-1}.

Consequently, we then used the high throughput 3D bioprinting protocol to print the multicellular bioink, and the constructs attained were subjected to further physical and biological characterization. As seen in figure 8(E), the morphology of 3D bioprinted microtumor can be seen. Live/dead assay results showcased that the cells in 3D microtumors were viable post-printing, and each construct carried sufficient cell density, as seen in the images in figure 8(F).

To highlight the presence of different cellular populations within the bioprinted cancer model, we investigated the presence of different surface markers of encapsulated MDA-MB-231 cancer cells, NIH-3T3 fibroblasts, and THP-1 monocytes in the cancer model. Figure 9 shows encapsulated cells in a printed model cultured until day 7 timepoint. It was observed that cancer cell clusters and the population of cells as single and isolated stellate-shaped entities coexisted through the culture period. We checked for multiple

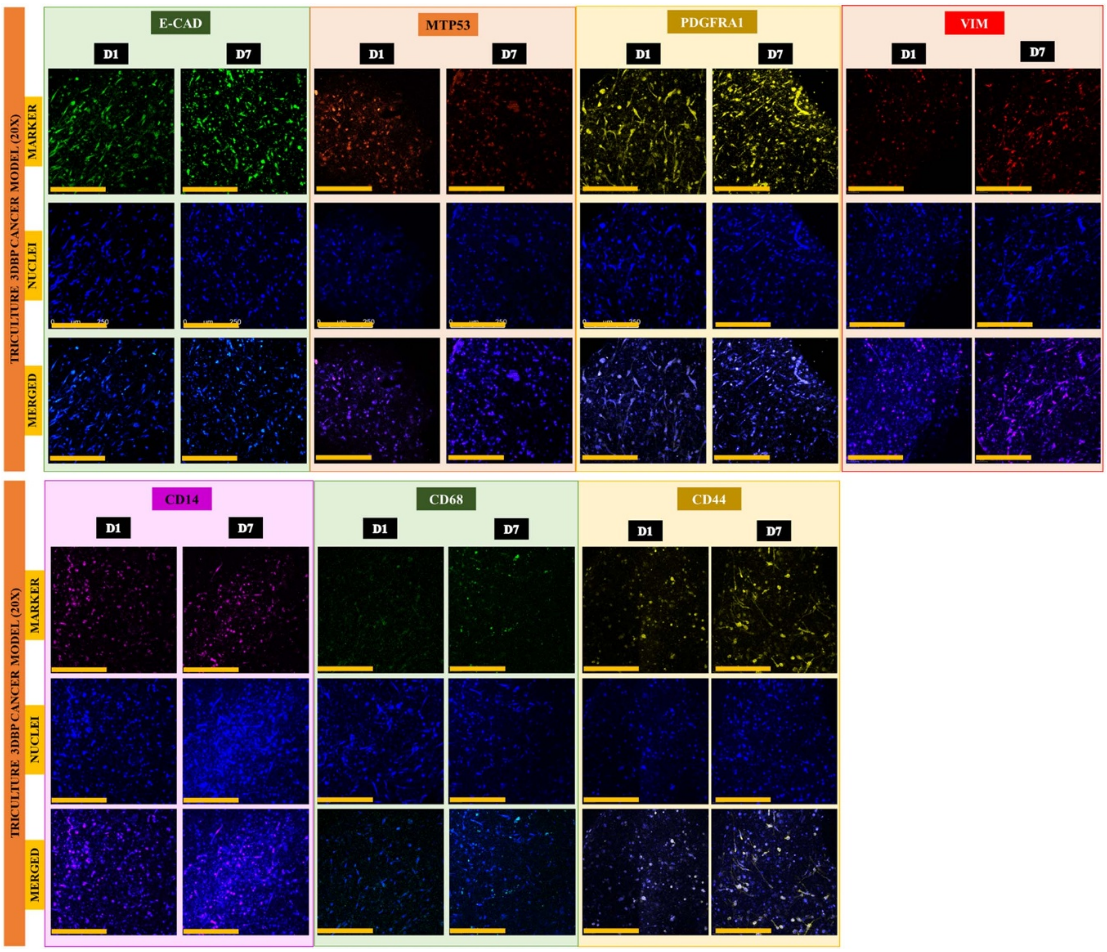

Figure 9. Cellular heterogeneity and functionality in immunocompetent triculture 3D bioprinted cancer model. Immunofluorescence images showing the expression of different functional markers in encapsulated cells in 3D microtumors on day 7 of culture within 3D bioprinted microtumors. (Scale bar = 250 μm).

cell surface markers like E-Cadherin, mutant p53, PDGFRα, Vimentin, CD14, CD68, and CD44.

E-cadherin is a cell surface receptor expressed at the sites of cell-to-cell adhesion and typically has a low to no expression level in cancer cells as well as monocytes, which are suspension cells. There was a marked increase in expression because of fibroblasts and cellular proliferation within the 3D model. MDA-MB-231 cancer cells have a mutant p53 genotype, while the fibroblasts and monocytes have a wild-type p53 genotype; hence, mutant p53 serves as a distinctive marker. The increased expression from day 1 to day 7 has been showcased in the respective panel. PDGFRα is a distinctive cell surface receptor used for fibroblasts that binds to PDGF ligands and is another marker for cells of mesenchymal origin. All the stretched fibroblasts and cancer-associated fibroblasts (CAFs) were seen to be highly expressing the marker across the culture period. Vimentin is a mesenchymal marker, typically expressed as intermediate filament proteins in cells. Expression for this marker was distinctively upregulated across culture periods, as seen in the panel.

CD14 is a cell surface marker for undifferentiated monocytes, serving as a distinctive marker against other cell populations in the model. The monocyte population can be seen encapsulated within the heterogeneous cancer model in the figure. CD68 was used to assess monocyte-to-macrophage differentiation as monocytes adhere to a surface. CD44 is a cell surface receptor that binds to hyaluronic acid and other components of ECM. In the panel of images, we observed a marked increase in expression across culture periods in the bioprinted model. The DAT hydrogel-based 3D bioprinted cancer model was able to support heterogeneous cellular populations and cell proliferation.

4. Discussion

Biomimetic cancer models with 3D tumor microenvironment and microarchitecture can serve as drug screening platforms, and their fabrication can be scaled up in a high throughput manner to develop a drug screening platform. This study presents a method of high throughput biofabrication of 3D breast cancer models using dECM-based hydrogel with the help of rapid 3D bioprinting technology. We chose DAT-based biomaterial to mimic breast cancer microenvironmental conditions, as mammary ducts and glands have adipose tissue as structural support [22–25]. Hence, we decellularized caprine origin adipose tissue and prepared a dECM-based hydrogel. We used a salt-based decellularization method modified from a previously known non-detergent-based method [27] compared to conventional detergent-based decellularization [20] to decellularize and delipidate adipose tissue. The non-detergent, salt-based decellularization routine used in our study proved highly effective, and the biochemical estimation, morphological, and histological results were in keeping with previously reported studies [28, 33, 41, 49, 65–67]. Effective removal of cellular debris and lipids associated with adipose tissue is necessary to prevent immunogenic reactions during cell culture-based experiments. Based on our research group's recent study [62], 62.385% $\pm$ 2.884% of hydroxyproline content and 80.606% $\pm$ 1.647% of sulfated GAGs are retained in our non-detergent DAT compared to native adipose tissue, while the DNA content for our decellularized tissue was below 50 ng mg^{-1} of lyophilized tissue.

We studied the gelation kinetics of our dECM-based hydrogel to understand the gel property of DAT hydrogel in response to temperature due to the thermoresponsive nature of collagen, the major protein in ECM. We observed that with a steady increase of complex modulus in temperature from 4 °C to 37 °C, indicating thermal crosslinking of collagen chains in dECM and the sol-to-gel transition that could allow for the formation of a 3D scaffold. A similar property has been previously observed for DAT-based hydrogels [20]. Our results also showcased a gradual and steady increase in complex modulus after 23 °C, indicating a crucial parameter that crosslinking is initiated well before physiological temperature. Hence, dECM-based hydrogels that have been pH adjusted need to be stored at cold temperatures and used within days of formulation of dECM pregel. We also assessed the stability of the crosslinked DAT hydrogel scaffold by its swelling index and found it to have supportive properties to absorb and aid the diffusion of nutrients within the hydrogel in a physiological environment. This indicated that cells encapsulated in a stable 3D hydrogel scaffold would have a cytocompatible microenvironment. Next, we assessed the flow behavior of DAT hydrogel by conducting a shear sweep experiment. Our results showcased that DAT hydrogel depicted shear thinning behavior and met the criteria for the viscosity profile required in a bioink [8, 68]. This was seen in previous studies where DAT hydrogel was used for bioprinting [20, 69]. We showcased that the extrusion-based 3D bioprinting process with DAT hydrogel did not hamper cell viability and that cancer cells continued to proliferate within the 3D bioprinted construct. dECM hydrogels have been proven to be an excellent natural biomaterial by providing tissue-specific native microenvironment cues. DAT hydrogel-based scaffolds have found their application for a variety of tissue engineering and clinical applications [20, 28, 31, 33, 41], including cancer research studies [42–47]. Previous studies have used DAT hydrogel along with a mechanically robust biomaterial as a framework or additional materials to formulate a bioink that would extrude a continuous filament [20, 41, 48]. We used cancer cell-laden bioink to bioprint a lattice structure to optimize the extrusion and showcase the printability of our DAT hydrogel without any additive material and compared the cellular morphology and viability with the conventional 2D TCP model. Recently, Chen *et al* used porcine DAT hydrogel mixed with GelMA as a viscosity modifier, and Blanco-Fernandez *et al* used two viscosity modifiers- GelMA and alginate to improve its printability to attain 3D bioprinted cancer model [48, 49]. Per the literature survey, we have not found preexisting studies using only DAT-based hydrogel as a single biomaterial bioink for developing 3D bioprinted breast cancer models. Additionally, we compared the gene expression profile for cancer cells grown in monolayer 2D culture with a bioprinted cancer model to understand how the three-dimensional microenvironment alters the genotypic profile of cancer cells in our dECM-based bioprinted model compared to conventional *in vitro* cultured cancer cells. To study gene expression independent of cell number in a 2D cancer model versus a 3D bioprinted cancer model, during our gene expression experiment, at the step of preparation of cDNA, we kept the RNA concentration for all samples at a constant value of 25 ng ul^{-1}. This ensured that varying cell numbers due to dimensionality at the stage of RNA isolation did not impact the RNA concentration taken for rtPCR, as the same amount of RNA was used for 2D versus 3D test groups. We found a significant increase in expression levels of genes specific to cancer stemness [46, 70, 71] as well as cancer cell invasion and migration [53, 72] in 3D bioprinted samples compared to 2D samples, and this can be attributed to the fact that the 3D microenvironment is a much more physiologically relevant experimental condition. Similar altered gene expression in comparison to 2D cultured cells has also been observed in other 3D bioprinted models [48, 73–77].

We modified our bioprinting approach to fabricate 3D microtumors in a high-throughput manner adapted to a 96-well plate format. The use of bioprinting technology is particularly beneficial for diagnostic applications where a large number of 3D test samples are beneficial, such as personalized cancer organoids for exploring different therapeutic regimens. High throughput bioprinting for drug screening has previously been shown for drug screening applications; however, low-viscous biomaterials like gelatin, collagen, Col-Ma, and hyaluronic acid in combination were used, while our study used biomimetic dECM-based hydrogel [53, 54]. The high throughput approach allowed us to use minimal reagents and dECM-based bioink. The fabrication process took 7 min compared to manual pipetting in a 96-well plate, which takes approximately 15 min. For comparison, in our study, the total time taken for one 96-well plate to be printed entirely was 7 min, while Maloney *et al* took 8 min with one printer and 19 min with another printer to bioprint their high throughput model [54]. We then characterized printing parameters like nozzle size and printing pressure as they directly impact the size of bioprinted experimental samples. Previous studies using different bioprinting modalities have shown that tuning different parameters like pulse width [47] and acoustic pressure field [66] can change the high throughput droplet radius and size of 3D cell aggregates. For extrusion-based bioprinting, high pressure can lead to cellular damage. At the same time, too low of a printing pressure would lead to an issue in consistent bioink extrusion [68] with biomaterial like DAT hydrogel, and therefore we characterized the scaffold diameter across a range of extrusion pressures and nozzle sizes. Consistency in the size of 3D bioprinted microtumors is also vital for applications such as organoid preparation, mainly done by hand in practice [78, 79]. Therefore, we also explored the consistency of the diameter of 3D microtumor and printing speed with manual pipetting using a single-channel pipette, considering that the DAT hydrogel is a much more viscous material [20, 41, 48] than liquid cell suspension typically used for 3D tumor models like spheroids and organoids [80]. We found more variation in the diameter of 3D microtumors printed with smaller nozzle sizes as compared to bigger nozzle sizes, and this could be due to the deposition property of DAT hydrogel in response to extrusion pressure experienced more in the smaller nozzle as compared to the larger nozzle, leading to over-extrusion of material in certain scaffolds and spreading of deposited volume. We further characterized the height of the 3D bioprinted microtumor, and we observed the spreading of the bioprinted scaffold from edges, which led to a reduced height, and this flattening phenomenon was more observable in scaffold printed with 250 μm nozzle as compared to 330 μm nozzle. It has been suggested in previous literature that larger nozzle diameters should be used to reduce printing errors like calibration accuracy, and nozzle sizes with inner diameter above 200 μm should be used for much more reproducible results [64]. Volumetric and biological characterization was done based on previous experiments [8, 64]. We found that the high throughput bioprinting process was cell-supportive and that a bigger nozzle size led to the deposition of more volume of bioink, hence forming a larger bioprinted scaffold, which showed higher cellular viability compared to that within the bioprinted scaffolds printed with a smaller nozzle size. Based on the results from previous studies where various cellular populations were encapsulated in dECM-based bioink at lower temperatures [20, 49, 63, 81], a bioprinting temperature of 15 °C was set to keep adequate balance between preventing thermal gelation of collagen within the bioink and cell supportive temperature. The encapsulated cells are prevented from going into temperature shock due to cushioning effect provided by surrounding dECM polymer network as discussed before. Cells tend to recover post-printing process within hours and scaffolds show high cell viability for the duration of culture [20].

We wanted to use the high throughput 3D bioprinted cancer model as an *in vitro* drug screening platform that could be scaled up and fully automated to give consistent results. For this purpose, we tested two anti-cancer drugs on our bioprinted platform. Since it has been shown in previous studies that cancer cell viability is different in 2D monolayer model compared to 3D bioprinted model in response to chemotherapeutic agents [48, 49, 82, 83], therefore we kept 3D bioprinted models as untreated control for our study. Our drug assay results were in keeping with the results with dose-dependent cytotoxicity but chemoresistance at lower concentrations for anti-cancer drug treatment in 3D bioprinted cancer models, where a significant difference was seen in cytotoxicity results for cells in 2D culture versus 3D scaffolds [74, 84]. Additionally, it should be noted that due to the encapsulation of cancer cells in a 3D bioprinted cancer model, the presence of extracellular matrix around cancer cells provided a realistic microenvironmental condition to test the anti-cancer drugs, as shown elsewhere [12, 85]. The 3D matrix also provides a barrier to the diffusion of drugs [86, 87] and hence can be another reason why the cells were viable and showing chemoresistance. In the future, from a pharmaceutical perspective, a more in-depth analysis of the mechanistic action of anti-cancer drugs can be studied on this platform to conduct and study the combinatorial effects.

Gene expression analysis of anti-cancer drug treated high throughput 3D bioprinted microtumor showcased their cytotoxic effect at the genetic level. Oct4 and Sox2 are two pluripotency-associated transcription factors upregulated in cancer stem cells [70]. Within cancer cells, the upregulation of these genes can give rise to cancer stem cell phenotypes that can evade chemotherapeutic agents and provide chemoresistance [71]. On exposure to chemotherapeutic agents, we found that the gene expression levels were downregulated compared to untreated control but not absent, and the same expression pattern has been observed in a previous study [46]. Similarly, the two genes MMP2 and MMP9 are known ECM proteases [72] and upregulated during invasion and migration [88]. In our study, these genes were downregulated in the cancer cells in drug-treated 3D microtumors as compared to untreated control, but not absent. Other 3D bioprinted cancer models have also noted these gene expression profiles [53, 89]. While normal healthy cells have a high p63 expression, the tumor suppressor gene belonging to the p53 family has two isoforms, TAp63 and ΔNp63 [90]. Functionally, in normal healthy cells, it is known to be a metastasis suppressor gene and share common downstream target genes with p53 that induce apoptosis [91, 92]. Within breast cancer cells, specifically MDA-MB-231 cells with mutated p53, due to conformational change in wild-type p53 structure, there is an introduction to weak binding of mutated p53 with p63. This leads to either no or low p63 expression in cancer cells, indicating increased invasion and migration capacity of cancer cells [93, 94]. The p53 reactivator, PRIMA-1^{Met,} works as an agent for restoring p53 functions; hence, p63 would not remain bound to mutant p53 [95]. In our expression analysis study, we could see that there was some level of expression of p63 in PRIMA-1^{Met} treated samples compared to untreated samples. In a previous study where COTI-2 was used as a p53 reactivator, there was an increased p63 and p73 expression in treated mutant-p53 TNBC cells grown in 2D monoculture conditions [96]. Similarly, within samples treated with 5-FU as well as combinatorial doses of both the anti-cancer drugs, there was only a non-significant increase in p63 expression, which is phenotypically reflected as loss of invasion and migration capacity of cancer cells after being exposed to anti-cancer drugs. A major gene involved in cell cycle regulation is p21, and one of the genes regulating it is p53. In normal cells, p21 can be induced due to varying conditions, such as DNA damage or cellular stress response, which can cause cell cycle arrest and induce apoptosis. However, p21 has a low expression in cancer cells, as indicated by unregulated cell division in tumors [91]. Our expression analysis showed that p21 expression in drug-treated samples was highly upregulated compared to the untreated cancer model, indicating apoptosis. Since PRIMA-1^{Met} leads to the restoration of p53 and transactivation of downstream genes, including p21, its expression also increased in PRIMA-1^{Met} treated samples, as seen in the previous study [95]. Due to the inherent cytotoxic effect of 5-FU, p21 expression increased as an indicator of the occurrence of apoptosis [97]. For combinatorial treatment, p21 expression increased more noticeably with increased PRIMA-1^{Met} concentrations.

We further increased the complexity of the bioprinted model to showcase the potential of using high throughput 3D bioprinting for heterogeneous tumor cell populations in constructs as seen *in vivo* or in patient tumor samples, which can be an important indicator of future studies in the field of personalized cancer models. We optimized heterogeneous, high-density, multicell-laden bioink extrudability by printing lattice-shaped structures. We conducted a high throughput bioprinting and observed post-printing cellular viability with cells uniformly dispersed across the printed 3D microtumors. Our study has incorporated all the cells in the same biomaterial within the same construct, rather than separate zones, to mimic the immune cell infiltrated tumor microenvironment and the presence of native ECM microarchitecture. Furthermore, we presented the expression profile for different surface markers seen within the 3D bioprinted breast cancer model to study the cellular organization within our model. MDA-MB-231 cells are mutant p53 TNBC cell lines, and we stained the cells with mutant-p53 specific antibody to showcase their distinct presence from other wild-type p53 cells, and an increase in immunofluorescence was seen with an increase in cancer cell population as previously noted for PAB240 antibody [96]. Cancer cells undergo epithelial-to-mesenchymal transition to promote, and two essential markers for this phenomenon are E-cadherin and Vimentin [98–100]. MDA-MB-231 cells usually have a low expression of E-cadherin as it is a non-mammosphere-forming cell line [101]. Vimentin expression is upregulated during EMT and known to be expressed in aggressive cancers; within our model, the expression of Vimentin increased while E-cadherin decreased over the culture period, indicating increased EMT activity [102–104]. Vimentin is also expressed in fibroblasts in native stromal microenvironment [100]. Within the tumor microenvironment, cancer cells interact with different cells present in stroma, including fibroblasts [105]. Cancer cells are known to recruit these fibroblasts, and a new phenotype of CAFs is seen [106, 107]. PDGFRα are surface receptors that bind to platelet-derived growth factors and are expressed across different cell types, including CAFs [108]. We saw its expression in the stromal cells present in our 3D microtumors. Previous studies have studied the interactions of breast cancer cells and fibroblast in different coculture studies, including 3D spheroids and within 3D bioprinted models [109–113]. CD44 surface expression was also seen increasing across the culture period and is known to correlate with breast cancer cell stemness, and we were able to see a homogenous distribution of cells expressing CD44 in our bioprinted model [114–117]. It has been noticed elsewhere that in response to shear forces, the expression of CD44 isoform increases due to the metastatic potential of MDA-MB-231 cells, and we concur that in our model, MDA-MB-231 cells were exposed to shear forces during the extrusion printing process and it might be one of the reasons of an increased expression pattern [118]. The tumor microenvironment is a heterogeneous cell ecosystem with a large portion of the non-cancerous cell population in the form of immune cells [119, 120]. CD14 is a marker for undifferentiated monocytes, and we could see the presence of monocytes in 3D bioprinted microtumors in our model. Previous studies have cocultured MDA-MB-231 cells with THP-1 cells in monoculture conditions and 3D hydrogels and introduced external factors to induce differentiation of the monocytes [121, 122]. CD68 is a monocyte-to-macrophage differentiation marker often used to distinguish differentiated cells from undifferentiated THP-1 monocytes [121, 122], and we observed its expressions within the 3D microtumors with increasing culture period in our study. This can be attributed to tumor microenvironmental conditions slowly differentiating the THP-1 monocytes toward tumor-associated macrophage lineage [123]. It has previously been reported that in TNBC patient samples, there is a marked presence of M2-like macrophages, which correlates with high immune responses and low survival rates [124]. Tumor-associated macrophages and exhausted T cell populations, majorly immunosuppressive in function, have been identified in breast cancer samples using transcriptomic analysis [125].

In summary, we addressed the issue of scaling up biofabrication of dECM-based 3D cancer models for rapid drug screening and downstream experimental analysis by considering tumor microenvironmental cues like the presence of 3D ECM structure and heterogeneous cell population. This high throughput biofabrication of DAT-based heterogeneous 3D cancer *in vitro* model has a future application in personalized cancer model development and drug screening application.

5. Conclusion and future perspective

The escalating demand for three-dimensional physiologically relevant disease models necessitates innovative testing platforms to circumvent the need for large-scale animal testing. Among these platforms, 3D bioprinting stands out as a game-changer, enabling the production of highly reproducible prototypes with remarkable efficiency. Particularly crucial is the development of a high throughput platform to meet the demands of drug screening assays and toxicity studies, enhancing sample size and quality for more robust downstream analysis and reliable data generation.

In our study, we harnessed discarded white adipose tissue to formulate a bioink for 3D bioprinting, paving the way for the creation of a TNBC model and the establishment of a high throughput platform. Through meticulous optimization, we achieved effective decellularization of the adipose tissue-based hydrogel using a non-ECM degrading protocol, ensuring cost-effective sourcing compared to commercially available alternatives. By innovating traditional 3D printing approaches into a high throughput protocol, we successfully biofabricated 3D microtumors in record time, under 7 min, for drug screening applications involving potent anti-cancer drugs like 5-FU and PRIMA-1^{Met}. This approach not only minimized biomaterial and reagent consumption but also maximized experimental model output for downstream analysis. Furthermore, we demonstrated the capability of our platform to biofabricate physiologically relevant 3D tumor models by incorporating fibroblasts and undifferentiated monocytes, thus achieving immunocompetent tumor models. This showcases the versatility and potential of 3D bioprinting technology in rapidly generating user-friendly, reproducible, and sophisticated drug screening platforms.

Our study underscores the transformative potential of DAT-based hydrogel in developing biomimetic heterogeneous breast cancer models with complex microarchitectures and biophysical cues. Looking ahead, the implications of this research extend to the biofabrication of patient-specific 3D bioprinted cancer models, promising enhanced screening capabilities and the tailored development of personalized therapeutics.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

Conflict of interest

The authors declare no conflict of interest. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Ethical statement

No human subjects, human data or tissue were used for this study, and no animal testing was conducted in this study too.

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